Superior University

Department Of Software Engineering

Classes Explanation

Project Name :

Car Rental System

Subject :

Object Oriented Programming

Section :

BSSE-2B

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1. Structure: CustomerInfo

A **struct** that stores all customer and booking information:

Personal Details: name, father's name, age, ID, phone number, address, occupation

Booking Details: car names, rental rates, booking dates, number of days, number of cars

Journey Information: pickup/drop locations, distance (km)

Payment Information: total amount, final amount after discount

Feedback: customer feedback

Static Instance: info maintains data throughout the program

2. Base Abstract Class

The abstract base class provides core functionality and defines the interface:

Key Components:

Data Members:

Booking options (opt, choice, Caroption, etc.)

Customer details (name, address, phone, etc.)

Static total to store rental amount

Journey details (pickup, drop, km)

Constructors:

- **Parameterized Constructor** (base(char Y)):

Implements human verification process

Uses getch() for pause and Sleep() for delays

Validates if user is human before proceeding

- **Default Constructor:**

Basic initialization

Pure Virtual Function:

virtual void adminMenuLoop () = 0; enforces derived classes to implement admin menu functionality

Customer Menu Functionality:

customerMenuLoop() provides:

1. Car selection (normal/bridal)
2. Booking management
3. Profile updates
4. Policy viewing
5. Feedback submission
6. Uses switch-case for menu navigation
7. Implements file I/O to save receipts

Input Validation:

The function checks if the user's input is within the valid range (0-6). If not, it prompts the user to enter a valid option.

Main switch Case:

The function uses a switch statement to handle the user's choice:

Case 1: **Check Available Car**

Displays available cars and allows the user to select a car.

Uses nested switch cases to handle different types of cars (e.g., normal cars and bridal cars).

Case 2: **View Booking**

Displays the current booking details.

Case 3: **View Company Policy**

Displays the company policies.

Case 4: Update Profile Details

Allows the user to update their profile information.

Case 5: **Cancel Booking**

Prompts the user to confirm the cancellation and then cancels the booking.

Case 6: **Customer Feedback**

Allows the user to provide feedback.

Case 0: **Logout**

Logs the user out and exits the menu loop.

Nested switch Cases:

When the user selects "Check Available Car" (Case 1), a nested switch case is used to handle the selection of different types of cars:

Case 1: **Bridal Cars**

Displays a list of bridal cars and their details.

Prompts the user to select a car and enter booking details.

Case 2: **Normal Cars**

Displays a list of normal cars and their details.

Prompts the user to select a car and enter booking details.

if-else Conditions:

The function uses if-else conditions to validate user input and control the flow of the program. For example:

Checking if the user's input is within the valid range.

Validating car selection and booking details.

3. Derived Class: Admin

The admin class is derived from the abstract base class and is responsible for managing administrative functionalities within the Car Rental System.

Key Components:

1. Constructors:

Default Constructor:

Initializes the base class with a verification character, indicating the start of the admin functionalities.

Parameterized Constructor:

Allows for custom initialization and verification, ensuring that the admin is authenticated before accessing the system.

2. Overriding the Pure Virtual Function:

adminMenuLoop():

This function overrides the pure virtual function from the base class, providing the specific implementation for the admin menu. It is essential for defining how the admin interacts with the system.

3. Menu and Functionality:

The **admin** class provides a structured menu that allows the admin to perform various operations, including:

1. **Enter Customer Information:**

Admin can input and save customer details such as name, age, ID, and contact information.

2. **Car Selection for Rent:**

Admin can select cars available for rental, categorized into bridal and normal cars.

3. **Enter Journey Information:**

Admin can input details regarding the journey, including pickup and drop locations, as well as the distance.

4. **Generate Payment Receipt:**

Admin can generate and display payment receipts, including calculations for discounts based on distance traveled.

5. **Add Payment:**

Admin can process payments, either full or partial, and update the total amount due accordingly.

6. **Logout:**

Admin can log out of the system, exiting the menu loop and returning to the main login interface.

4. Switch Cases for Navigation:

The **adminMenuLoop**() function employs a switch-case structure to handle user selections:

Each case corresponds to a specific functionality, allowing the admin to navigate through the menu efficiently.

Case 1: Enter Customer Information

Prompts the admin to input customer details and saves the information.

Case 2: Car Selection for Rent

Displays options for selecting bridal or normal cars, allowing the admin to manage car rentals.

Case 3: Enter Journey Information

Collects journey-related data from the admin.

Case 4: Generate Payment Receipt

Calculates and displays the total amount due, including any applicable discounts.

Case 5: Add Payment

Processes payments and updates the total amount due.

Case 0: Logout

Logs the admin out of the system.

5. Input Validation:

The admin class includes input validation to ensure that the admin's selections are within the valid range. If an invalid option is selected, the system prompts the admin to enter a valid choice.

6. Nested Switch Cases:

Within the car selection functionality, nested switch cases are utilized to differentiate between bridal and normal cars:

Bridal Cars: Displays a list of available bridal cars and prompts the admin to select one.

Normal Cars: Displays a list of available normal cars and prompts the admin to select one.

7. If-Else Conditions:

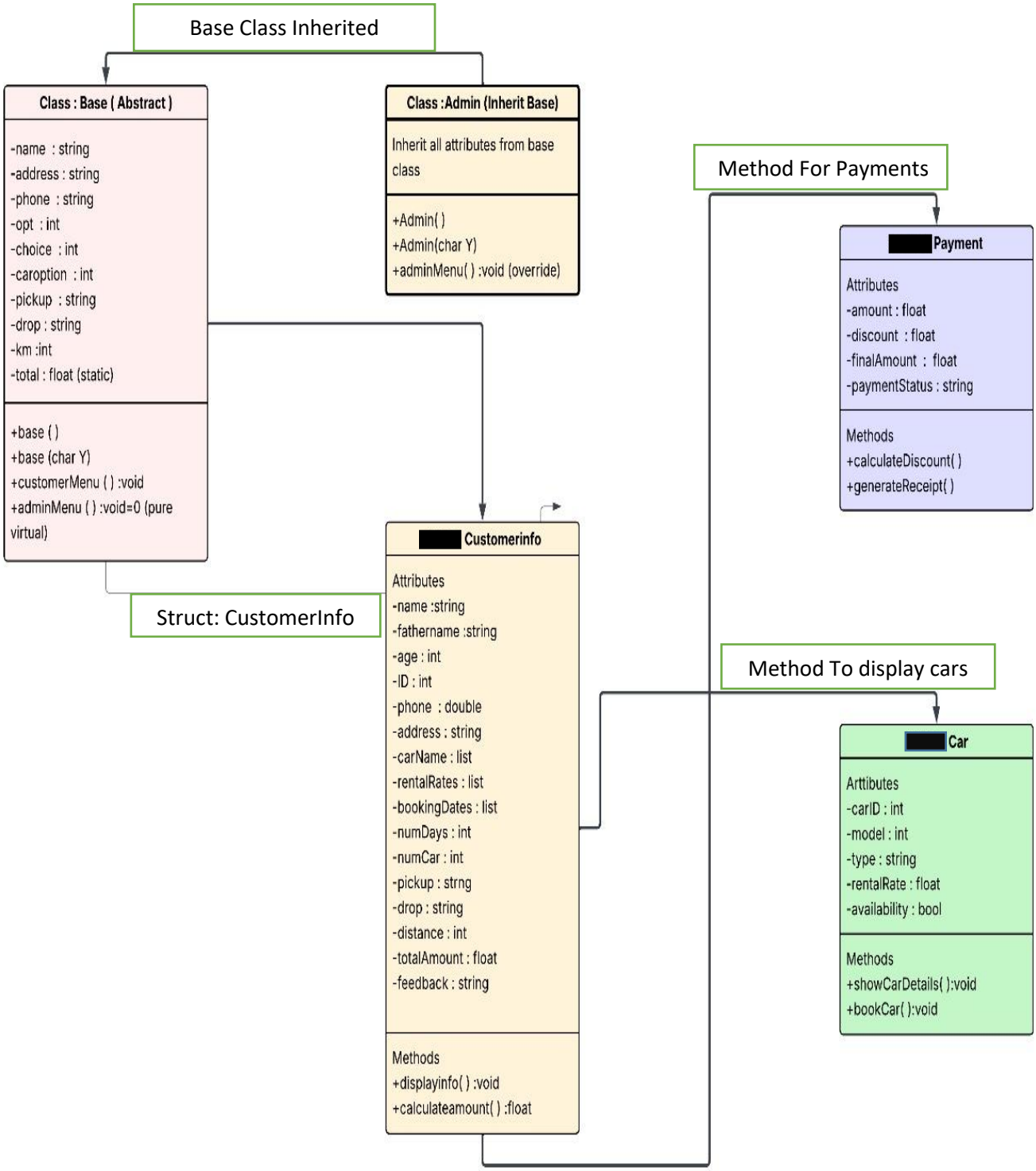
The admin class employs if-else conditions to validate user input and control the flow of the program. This includes:

Checking if the selected car is valid.

Validating booking details and ensuring that all required information is provided.

Concepts Used In Classes:

1. **Inheritance**:
- The `admin` class inherits from the abstract base class `base`, allowing it to reuse data members and member functions.
2. **Abstraction**:
- The `base` class is abstract due to the pure virtual function `adminMenuLoop()`, defining an interface for derived classes.
3. **Polymorphism**:
- The `adminMenuLoop()` function is overridden in the `admin` class, allowing dynamic method behavior when called through a base class pointer.
4. **Modularity**:
- The program is organized into classes based on responsibilities, such as handling admin functions and customer information.
5. **Dynamic Memory Allocation & Polymorphism (in Main)**:
- In `main()`, an `admin` object is created using dynamic memory allocation via a base class pointer, demonstrating dynamic dispatch.


Base Class Inherited
Class : Base (Abstract)
-name : string
-address : string
-phone : string
-opt : int
-choice : int
-caroption : int
-pickup : string
-drop : string
-km :int
-total : float (static)
+base ( )
+base (char Y)
+customerMenu ( ) :void
+adminMenu ( ) :void=0 (pure virtual)
Class :Admin (Inherit Base)
Inherit all attributes from base class
+Admin( )
+Admin(char Y)
+adminMenu( ) :void (override)
Method For Payments
[redacted] Payment
Attributes
-amount : float
-discount : float
-finalAmount : float
-paymentStatus : string
Methods
+calculateDiscount( )
+generateReceipt( )
Struct: CustomerInfo
[redacted] Customerinfo
Attributes
-name :string
-fathername :string
-age : int
-ID : int
-phone : double
-address : string
-carName : list
-rentalRates : list
-bookingDates : list
-numDays : int
-numCar : int
-pickup : strng
-drop : string
-distance : int
-totalAmount : float
-feedback : string
Methods
+displayinfo( ) :void
+calculateamount( ) :float
Method To display cars
[redacted] Car
Arttibutes
-carID : int
-model : int
-type : string
-rentalRate : float
-availability : bool
Methods
+showCarDetails( ):void
+bookCar( ):void